

Max Marks: 70

Time Allowed: 3 hours

Name: _____

ERP: _____

Question:	Short Questions	Tracing	Conditions	Arrays	Recursion	OOP	Total
Marks:	20	16	6	12	6	10	70
Score:							

Question 1: Short Questions 20 marks

- (a) [10 marks] For each of the following boolean expressions, state whether it is always **true** (T), always **false** (F), sometimes **true** and sometimes **false**/not enough information to tell (S), will result in a compiler error (CE), or will result in a run-time exception (RE). Assume there are no rounding errors and the variables **x** and **y** are **ints**.

i. `Double.parseDouble(3.0) == 3.0`

i. _____

ii. `3 / 2 == 1.5`

ii. _____

iii. `2 * x / 2 == x`

iii. _____

iv. `2(3) == 6`

iv. _____

v. `1 / 0 == 1.0 / 0.0`

v. _____

vi. `Math.sqrt(x) * Math.sqrt(x) == x`

vi. _____

vii. `x - Math.abs(x - y) != y`

vii. _____

viii. `1.0 + 2.0 == 3.0`

viii. _____

ix. `Double.parseDouble("3") == 3.0`

ix. _____

x. `x - y <= x`

x. _____

- (b) [10 marks] Identify 7 bugs in the program below that will prevent it from compiling or running. For each bug, give the line number and corrected line of code. Write your answers in the provided space.

We will accept any reasonable fix that allows the program to compile and run without error. You do not need to worry about the program's purpose.

```
1  public class LittleBuggy() {
2      public static void main(String args) {
3          int input = args[0];
4          if(input < 1) return;
5          double arr = new double[input];
6          int i;
7          for (int j = 0; j <= arr.length; j += 1) {
8              i = (i + j) % arr.length;
9              arr[j] = compute(i, j);
10             System.out.println("" + arr[j]);
11         }
12     }
13     public static String compute(int i, int j) {
14         if(i < j)
15             return "a";
16         if(i > j)
17             return "b";
18     }
19 }
```

Bug 1:

.....

Bug 2:

.....

Bug 3:

.....

Bug 4:

.....

Bug 5:

.....

Bug 6:

.....

Bug 7:

.....

Question 2: Tracing 16 marks

(a) [10 marks] Read the code below, then answer the questions in the provided space.

```

public class Playground {
    public static void main(String[] args) {
        int n = Integer.parseInt(args[0]);
        if (n == 1) {
            int d = slide(Integer.parseInt(args[1]),
                          Integer.parseInt(args[2]),
                          Integer.parseInt(args[3]));
            System.out.println((d + "-" + (d + 5 + "-" +
                                   ((d + 12) + "-" + d + ("-" + 83)));
        } else if (n == 2) {
            double d = seesaw(Integer.parseInt(args[1]));
            System.out.println(d);
        } else if (n == 3) {
            String d = sandbox();
            System.out.println(d);
        }
    }
    public static int slide(int a, int b, int c) {
        if (b < 1) return (a + c);
        a *= 2;
        return slide(a, b-1, c);
    }
    public static int seesaw(int a) {
        while (a < 81) a += 2;
        return a;
    }
    public static String sandbox() {
        String[] arr = {"N", "U", "3", "R", "F"};
        return arr[4] + arr[1] + arr[Integer.parseInt(arr[2]) -
                                   Integer.parseInt(arr[2])];
    }
}

```

- i. What does the command “java Playground 1 2 5 3” print?
.....
- ii. What does the command “java Playground 2 2” print?
.....
- iii. What does the command “java Playground 3” print?
.....

- iv. Describe in 20 words or less what `slide()` computes. You may assume that `a`, `b`, and `c` are all at least zero.

.....

- v. Describe in 20 words or less what `seesaw()` does. You may assume that `a` is at least zero.

.....

- (b) Given the following class, answer the questions below:

```
public class Fever {
    public static int restless(int[] arr) {
        int sum = 0;
        for (int i = 0; i < arr.length; i++)
            sum = sum + arr[i];
        return sum / arr.length;
    }
    public static void main(String[] args) {
        int[] intArgs = new int[args.length];
        for (int i = 0; i < args.length; i++)
            intArgs[i] = Integer.parseInt(args[i]);
        System.out.println(restless(intArgs));
    }
}
```

- i. [3 marks] Circle the statements below true or false

- α) TRUE FALSE will not compile because `restless()` is called inside `System.out.println`
 β) TRUE FALSE `intArgs[]` is `args[]` converted into an integer array
 γ) TRUE FALSE may crash with an `ArrayIndexOutOfBoundsException` error when run

- ii. [3 marks] Assuming that any errors you identified above are corrected, what will the program print out in each of the following cases? We will consider your answer correct as long as it shows you understand what the program will print out.

- α) % java Fever

.....

- β) % java Fever 1.3 2.5 7.1

.....

- γ) % java Fever 1 2 5

.....

Question 3: Conditions *6 marks*

Write a program that qualifies the user for a particular credit card. Users must meet the following criteria:

- Make at least 30,000 per year
- Their monthly rent payment must not be more than 5% of their total salary (i.e. 1,500 is the max monthly rent that they can pay while making 30,000 per year)
- However, if they own their home you do not need to take their monthly house payment into account

The program takes input from command-line arguments. The first argument is the yearly salary. The second argument is either “y” or “n” describing if the user own their home or not. If the the second argument is “n”, then there is 3rd argument describing the monthly rent. Examples:

```
% java CreditCardQualifier 25000 n 500
```

Sorry, you don't qualify. Your salary is too low

```
% java CreditCardQualifier 35000 n 1000
```

You qualify!

```
% java CreditCardQualifier 30000 y
```

You qualify!

```
% CreditCardQualifier 50000 n 5000
```

Sorry, you don't qualify. Your rent is too high

[illegible]

Question 4: Arrays *12 marks*

- (a) [6 marks] Write a static method that takes as input an array of **doubles** sorted in descending order and a new **double**, and returns a new array that contains all **doubles** and is sorted in descending order. Examples:

$$\{8, 4, 3\}, 7 \rightarrow \{8, 7, 4, 3\}$$
$$\{9, 5, 3, 2\}, 10 \rightarrow \{10, 9, 5, 3, 2\}$$
$$\{7, 3\}, 1 \rightarrow \{7, 3, 1\}$$
[illegible]

- (b) [6 marks] Write a static method that takes as input an array of **ints** and returns **true** if the array contains two numbers whose values are consecutive. Examples:

$\{3, 8, 4\} \rightarrow \text{true}$ (3 and 4)

Question 5: Recursion *6 marks*

Write a recursive function that, given two strings, returns whether the first string is a subsequence of the second. For example, given `hac` and `cathartici`, you should return **true**, but given `bat` and `table`, you should return **false**.

Question 6: OOP 10 marks

A carnival ride has a name, has a maximum number of riders it can hold, can be for children or adults, and has a total number of tickets that have been collected from riders.

- (a) Complete the following instance variables declarations for a class **CarnivalRide** that models a carnival ride:

```
private _____ name;

private _____ capacity;

private _____ isKiddieRide;

private _____ numTickets;
```

- (b) Write a constructor for the **CarnivalRide** class that initializes a carnival ride with the name, the maximum number of riders it can hold, and whether it is for children or not as specified by the parameters. Initially, no tickets have been collected.

```
.....
.....
.....
```

- (c) Write an method **getTotal** for this class that returns the total number of tickets collected for the carnival ride.

```
.....
.....
.....
```

- (d) Write a method **loadRiders** for this class the loads riders onto the ride for the next ride. The number of riders loaded is specified in the parameter. If the number of riders is negative do not load any riders. If the ride is for children, the charge for a ride is 1 ticket, otherwise it is 3 tickets.

```
.....
.....
.....
```

- (e) In a different class than the **CarnivalRide** class, show how to declare a variable **dumbo** and assign to it a reference to a kiddie carnival ride object for a “Dragon” ride that can hold 16 children.

```
.....
.....
.....
```